

RESOURCE

Stress combinations and their interactions in plants database: a one-stop resource on combined stress responses in plants

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SUMMARY

We have developed a compendium and interactive platform, named Stress Combinations and their Interactions in Plants Database (SCIPDb; <http://www.nipgr.ac.in/scipdb.php>), which offers information on morpho-physio-biochemical (phenome) and molecular (transcriptome and metabolome) responses of plants to different stress combinations. SCIPDb is a plant stress informatics hub for data mining on phenome, transcriptome, trait-gene ontology, and data-driven research for advancing mechanistic understanding of combined stress biology. We analyzed global phenome data from 939 studies to delineate the effects of various stress combinations on yield in major crops and found that yield was substantially affected under abiotic–abiotic stresses. Transcriptome datasets from 36 studies hosted in SCIPDb identified novel genes, whose roles have not been earlier established in combined stress. Integretome analysis under combined drought–heat stress pinpointed carbohydrate, amino acid, and energy metabolism pathways as the crucial metabolic, proteomic, and transcriptional components in plant tolerance to combined stress. These examples illustrate the application of SCIPDb in identifying novel genes and pathways involved in combined stress tolerance. Further, we showed the application of this database in identifying novel candidate genes and pathways for combined drought and pathogen stress tolerance. To our knowledge, SCIPDb is the only publicly available platform offering combined stress-specific omics big data visualization tools, such as an interactive scrollbar, stress matrix, radial tree, global distribution map, meta-phenome analysis, search, BLAST, transcript expression pattern table, Manhattan plot, and co-expression network. These tools facilitate a better understanding of the mechanisms underlying plant responses to combined stresses.

Keywords: biocuration, climate resilience, data visualization, plant stress informatics hub, integretome, pathway enrichment, phenome, transcriptome.

INTRODUCTION

Abiotic and biotic stresses are the major deterrents to the achievement of global food security, necessitating the urgency to develop better-adapted crops (IPCC, 2022; Mittler & Blumwald, 2010). Plants are often exposed to combinations of stresses during their life cycle, and increasing evidence highlights that combined stress is a more potent and realistic threat to plant growth and productivity than individual stresses (Ahuja et al., 2010; Atkinson & Urwin, 2012; Desaint et al., 2021; Sinha et al., 2021; Zandalinas, Fritschi, et al., 2021; Zandalinas, Sengupta, et al., 2021). Considerable information on plant stress has

accumulated over the years, but current understanding of the physiological and molecular responses of plants to combined stress is inadequate (Mahalingam et al., 2021; Pandey et al., 2017; Zandalinas, Fritschi, et al., 2020; Zandalinas & Mittler, 2022). Combined stress studies, although under-represented compared to individual stress studies, entail voluminous and highly complex information on molecular mechanisms of plant defense (Cohen et al., 2021; Zandalinas, Fichman, et al., 2020; Zandalinas, Fritschi, et al., 2021).

A plant perceives combined stress as a new state of stress, the physiological and molecular response of plant to combined stress is often entirely different from the

response under the individual stresses. Therefore, the adaptation strategies to stress combinations are based on the interaction between the physiological and molecular responses simultaneously triggered by each stress entity independently (Gupta et al., 2016; Lopez-Delacalle et al., 2021; Pandey et al., 2017; Rizhsky et al., 2004). The outcome of such interactions may be “positive” or “negative,” wherein combined stress causes less or more damage, respectively, than that of the individual stresses (Figure S1). The outcome also depends on factors like plant age, genotype, stress intensity, duration of the stresses, order of stress perceived by the plant, and time of initiation of subsequent stress since the first stress, all of which make combined stress more complex to understand (Choudhary & Senthil-Kumar, 2022; Mittler, 2006; Pandey et al., 2015; Sinha et al., 2022; Zandalinas, Sengupta, et al., 2021). Further, reproductive tissues show more sensitivity towards some stress combination like heat and drought which leads to greater yield reduction (Sinha et al., 2022). These responses are mediated by switching on specific pathways and processes that are unique, specific, and sometimes even contrasting from the individual stress responses. Plants also exhibit shared responses common to individual and combined stresses (Suzuki et al., 2014; Zhang & Sonnewald, 2017).

Thus, to better comprehend the complexities of combined stress responses and fill existing gaps, there is a pressing need for a pertinent database. There is no

database or platform solely dedicated to combined stress. TOMRES (<https://www.tomres.eu/>) and Stress Combination: A New Field in Molecular Stress Research by the University of North Texas (<http://biology.unt.edu/stresscombination/>) are two web resources on combined stress in specific plants, besides individual stress databases such as STIFDB2, QlicRice, and the Arabidopsis stress-responsive gene database (Borkotoky et al., 2013; Naika et al., 2013; Smita et al., 2011). Although these resources are immensely useful in their own way, they do not contain updated information on stress combinations. The Stress Combinations and their Interactions in Plants database (SCIPDb; <http://www.nipgr.ac.in/scipdb.php>) developed by us is a user-friendly platform providing options to browse, search, analyze, and download data for various stress combinations studied to date. SCIPDb provides researchers easy access to combined stress-related information and tools for extracting need-based, specific information.

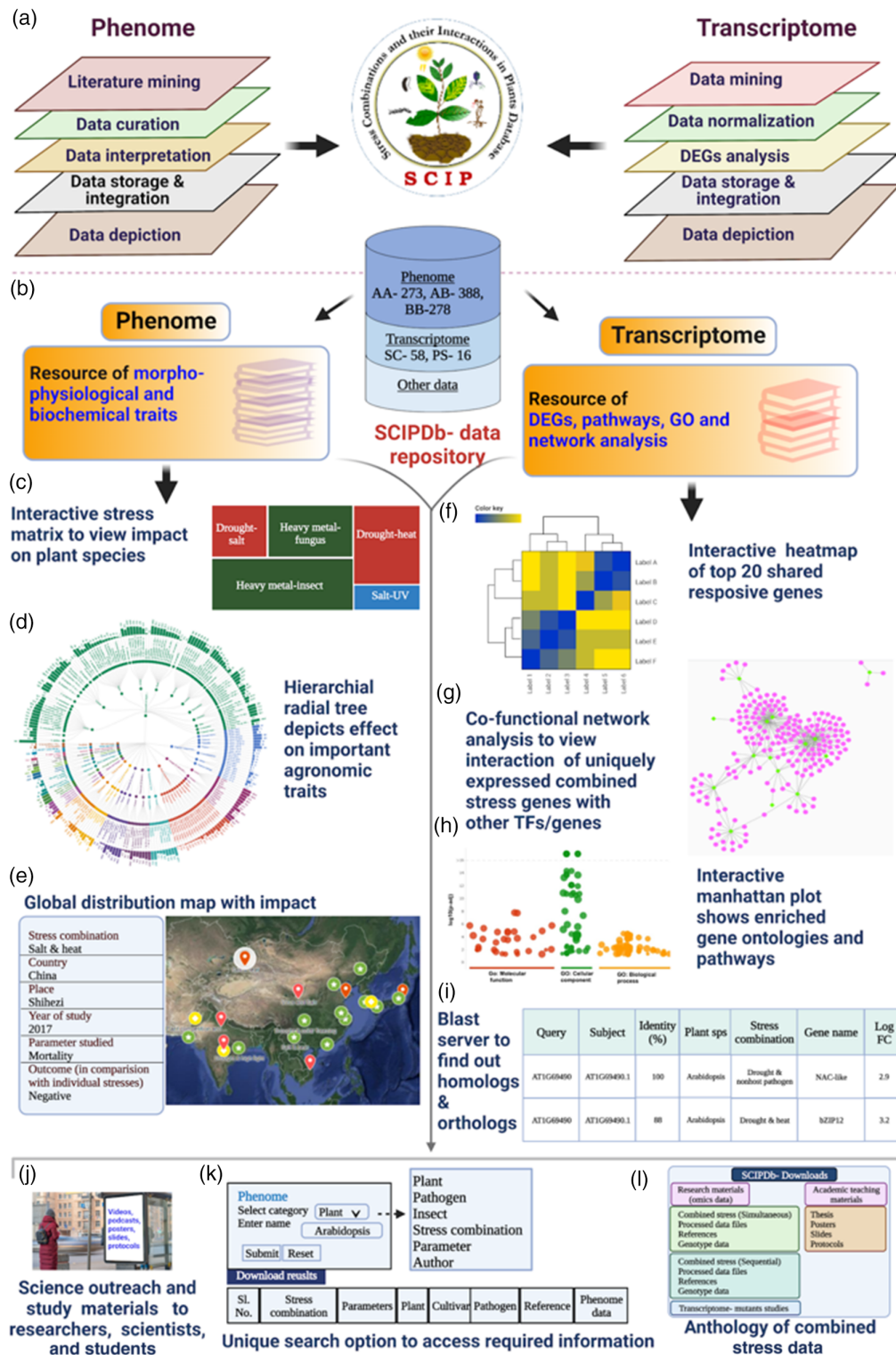
RESULTS AND DISCUSSION

SCIPDb and its key features

SCIPDb is a comprehensive collection of morphological, physiological, biochemical, and transcriptomic data on combined stresses published to date, systematically analyzed and presented in an easy-to-use interactive database and web server (Figure 1a,b; Figure S2). Currently, SCIPDb

Figure 1. Outline of the Stress Combinations and their Interactions in Plants Database (SCIPDb), indicating its key features and applications.

- The upper panel shows the steps involved in data mining, curation, analysis, and integration of phenome and transcriptome data into SCIPDb.
- The lower panel shows the key features and applications offered to the users in the phenome and transcriptome sections. Orange boxes indicate the two major data sets hosted in SCIPDb.
- The interactive stress matrix shows the net impact of the interaction between the stresses. The net impact of combined stress was determined by analyzing the percent reduction in plant growth, yield, and physiological traits. Three possible interactions, namely, positive (less damage under combined stress), negative (greater damage under combined stress), and others (equal damage under combined and individual stress), are depicted in green, red, and blue boxes, respectively. A stress combination is classified as positive, negative, or others based on the maximum number of studies in a particular interaction. The size of the box indicates the number of studies showing a particular interaction, i.e., a bigger size represents a greater number of studies.
- The radial tree depicts the effect of individual and combined stresses on various traits in different plant species arranged in hierarchical order (starting from most to least damage). The parameters considered for developing radial trees were growth, yield, physiological, and pathogen-associated parameters. To normalize the data, percent change over control or percent change over individual stresses (in the case of pathogen-associated traits) is calculated for each trait and presented.
- The interactive global map provides information on the global distribution of combined stresses and their effect on morpho-physiological, biochemical, and pathogen-associated traits. The map was generated using the geographic coordinates of the locations where the studies were conducted.
- An interactive heatmap enables users to visualize the gene expression profile of the top 20 differentially expressed genes (DEGs) for a particular transcriptome.
- A co-functional network depicts the correlation of the top 20 DEGs in the form of an interactive network. Nodes represent correlates (pink) of top 20 DEGs (green). Edges represent a log-likelihood score. The co-functional network allows the user to interact with the graph, and it includes all the required gestures, including pinch-to-zoom, box selection, panning, etc., to access other metadata for each node and edge.
- An interactive Manhattan plot depicts functional profiling of DEGs using various kinds of biological evidence, including Gene Ontology terms, biological pathways, and regulatory DNA elements. The X-axis represents functional terms grouped and color-coded by data sources, while the Y-axis shows the adjusted enrichment *P*-values in negative log₁₀ scales.
- BLAST server to find potential homologs and orthologs in SCIPDb.
- SCIPDb hosts science outreach materials like posters, slides, videos, and podcasts related to combined stress, which will be useful for students, researchers, and scientists working in the area of combined stress.
- The unique keyword-based search option helps to access all combined stress-related data with a single click. Searches can be performed using keywords like plant name, pathogen, insect, name of combined stress, gene ID, and gene name.
- The download section provides processed phenome and transcriptome data and a reference list of combined stress articles hosted in SCIPDb. It also provides a link to mutant transcriptome studies and diverse resources related to combined stress. Numbers within the cylinder indicate the total number of articles curated and presented in the phenome and the total number of stress combinations covered under transcriptome. AA, abiotic–abiotic stress; AB, abiotic–biotic stress; BB, biotic–biotic stress; SC, stress combinations; PS, plant species. The figure was created with [BioRender.com](https://www.biorender.com).



hosts phenome data curated from 939 studies, covering 123 stress combinations, 118 plant species, 283 pathogenic agents (including bacteria, fungi, oomycetes, nematodes, viruses, mycoplasmas, viroids, and insects), and 7 weed

species (Figure 1). From the analysis of the phenome data, 107 agronomic traits affected by various stress combinations were identified. Of these, 45 traits were mapped to the identified Trait Ontology (TO) terms. Twenty traits

among the 45 were related to plant morphology and yield; 11 were related to plant physiology; and 15 were biochemical traits corresponding to changes in enzymes, metabolites, and hormone levels. These traits can be targeted for trait-based breeding programs to develop combined stress-tolerant crops. Further, gene-to-TO relationships (Pan et al., 2019) can be derived to decipher the genome-to-phenome relationships under combined stress. The transcriptome data hosted in SCIPDb is from 36 studies available in the public domain thus far, representing 58 stress combinations and 16 plant species (Figure S3).

Phenomics

Systematically analyzed phenomes are presented as data pages based on plant species (Figure S4). Data pages are arranged and presented based on plant species under each stress combination. The whole data pages can be divided into two parts, first part explains the overall impact of combined stress in comparison with individual stresses and second part provides information on interaction between the stresses at plant interface under combined stress (Figure S4). For a holistic view of trends in the analyzed phenome data, interactive visualizations such as combined stress matrices, radial trees, and global combined stress distribution maps have been provided (Figure 1). The interactive stress matrices in SCIPDb show the overall impact of different stress combinations on various plant species (Figure 1c; http://223.31.159.3/plant_complete/matrix.html). These visualizations will aid in deciphering and distilling comprehensive overviews of stress combinations across plant species, unlike from individual stresses alone. Among the 123 stress combinations, 69 combinations showed a negative impact on plant growth and productivity (Figure 2; Figure S5). Twenty stress combinations showed a positive impact on plants, many of them belonging to the abiotic–biotic stress category. No combination among the 41 abiotic–abiotic stress combinations studied was observed to exhibit a net positive interaction, highlighting abiotic stress combinations as significant threats to plant yield (Figure S5, http://223.31.159.3/plant_complete/matrix.html). This trend was also predicted in earlier studies (Zandalinas et al., 2022; Zandalinas, Sengupta, et al., 2021). In 12 stress combinations, an equal number of studies reported both positive and negative impacts of combined stress on plants, with the majority belonging to the abiotic–biotic stress category (10 combinations) (Figure S5). To decipher the impact of stress combinations on growth, yield, and physiological and pathogen-associated traits in various plant species, data were visualized in the form of a radial tree (Figure 1d). Our analysis of the different abiotic and biotic stress combinations reveals many pathogen infections that are aggravated under several concurrent abiotic stresses. It also reflects a number of pests–pathogen complexes that can

pose a challenge to agricultural productivity. A global combined stress distribution map, another feature of SCIPDb, shows the prevalence of particular stress combinations in a locality with their impact on crop growth (Figure 1e). Knowledge of the occurrence of important stress combinations based on this interactive geographical distribution map can assist researchers in identifying agronomically relevant stress combinations. Our analysis of studies published from 1952 to 2021 showed an increasing trend in combined stress research, most of which were from the Americas, Asia, and Europe, particularly from the last decade (Figure S6a,b). Thus, the increase in the occurrence of combined stresses in these areas is deepening crop losses.

Transcriptomics

Transcriptome data were analyzed and presented as interactive bootstrap tables, which enlist differentially expressed genes (DEGs) and their associated metadata in the form of KEGG pathways and genes (Figure S7). Cross-references to important resources are provided to enable users to acquire more information directly. To further visualize the high-dimensional transcriptome data, DEG tables for common (shared genes between combined and individual stresses) category has been linked to interactive heatmaps and Venn diagrams while for unique (genes uniquely expressed under particular combined stress) category we have added co-functional networks and Manhattan plots (Figure 1f–h). SCIPDb hosts co-functional networks for the top differentially expressed unique genes under multiple combined stresses. It provides speculative evidence about the genes that are co-regulated and thus might share a similar biological function or act together to control a specific phenotype. The functional annotation of DEGs acts as a key resource to elucidate the biological processes, functions, and pathways controlling various combined stresses in plants. Gene Ontology annotations provided in the form of Manhattan plots can be used to visualize enriched biological processes, molecular functions, and cellular components and pathways.

Metabolomics

Metabolome datasets integrated in SCIPDb is from 6 different studies available in the public domain thus far, from 4 plant species. Out of this, drought and heat stress combination is the most studied stress combination represented in 5 studies. The presented metabolomic data encompasses a comprehensive compilation of metabolites, accompanied by their respective log fold change and absolute fold change in comparison to the control group. These values were derived through meticulous in-house analysis of the amalgamated stress metabolome, readily accessible within the public domain. Within this section, users are afforded the opportunity to effortlessly navigate through

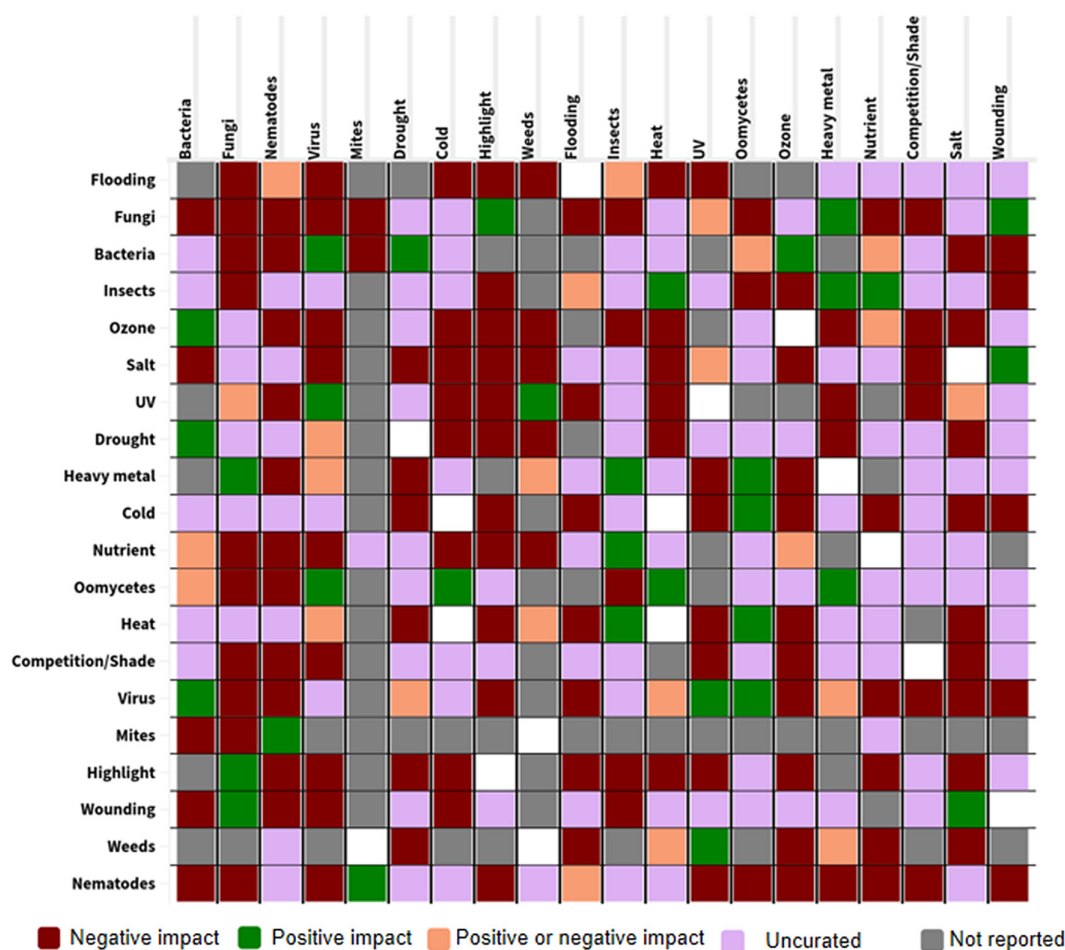


Figure 2. Stress matrix depicting the various stress combinations that affect plants.

Out of the 157 potential stress combinations identified through literature mining, SCIPDB presents curated phenomic datasets from 123 stress combinations. The colors in the matrix, represented here, depict the impact of the stress combinations based on findings of available studies as analyzed in the database. Dark maroon box- Negative impact; majority of studies indicate a negative impact of stress combination, i.e. plants under these combined stresses are affected to a greater extent as compared to individual stresses. Green box- Positive impact; majority of the studies indicate a positive impact of the stress combination implying plants under these combined stresses are less or equally affected as compared to one or both the individual stresses. Orange box- Positive or negative impact; equal number of studies with positive or negative impact of stress combination. Purple box- Others/reported/uncurated; indicating the stress combinations that are known to affect plants but are not curated in the database. Gray box- Not reported; depicting stress combinations that are not yet reported or are less likely to exist. White box represents stress combinations of same stresses that are practically not feasible. The X and Y axes indicate different abiotic (drought, cold highlight, flooding, heat, UV, ozone, heavy metal, nutrient [deficiency or toxicity] and salt) and biotic (bacteria, fungi, oomycetes, virus, mites, nematodes, insects and weeds) stresses. The details of rule between the combination's relationships are detailed in the materials and methods and also in the database (under methodology).

the metabolomics data. This is facilitated by a user-friendly interface, which utilizes dropdown menus for the selection of specific plants and their corresponding stress combinations. Moreover, to cater to the diverse needs of users, the provision for downloading the complete dataset resulting from user selections is also made available, streamlining access to the entirety of the data for further exploration and analysis.

Additional features of SCIPDB

A large fraction of genes in non-model plant species remains uncharacterized, which means that they lack functional annotation. Prioritizing candidate genes without any

functional evidence in such species is challenging. The standalone BLAST server integrated with SCIPDB provides an option to query the database with batch nucleotide or protein sequences and will help users identify genes related to combined stress in the genomes of the ever-increasing repertoire of newly sequenced crop species (Figure 1i). SCIPDB hosts several videos, slides, podcasts, and protocols related to combined stress, making it a potential outreach portal to promote scientific communication and education (<https://www.youtube.com/channel/UCamQH8rGtNMBvjuyNiSVyCw>) (Figure 1j). A unique keyword-based search option allows a user to mine desired information from both phenome and transcriptome

datasets (Figure 1k). SCIPDb datasets are hosted on a local FTP server, allowing users to download all curated phenomes, genotypes, transcriptomes, and references locally with just a few clicks. A user-defined download can also be done through specific sections of the database. These datasets can be further used for other downstream analyses to clearly grasp plant responses to combined stresses (Figure 1l). SCIPDb also encourages users to submit their data to the web portal to promote two-way communication and ultimately contribute to making the database a dynamic, robust, and single-stop platform for disseminating novel findings on combined stresses.

The interactive network developed by global combined stress transcriptome profiling and pathway enrichment analysis in *Arabidopsis* depicts the common and unique pathways between major combined stress categories hosted under the transcriptome visualization section. The “Applications” section hosts several case studies, which can help users understand how to use the diverse datasets hosted in SCIPDb to answer various biological questions about combined stress. The “References and links” section provides access to complete references of the research articles used in developing the data page, along with other related articles (review, thesis, and report categories). A meta-phenome presents a combined trend of the net impact of stress combinations on plant performance after analyzing all the studies reported for a specific crop for a particular stress combination. Overall, these important features and tools in SCIPDb provide comprehensive information on each stress combination and can help identify the most prominent stress combination in a specific crop affecting polygenic traits like growth and yield.

Key findings from the database

Impact of combined stress on yield and yield-attributing traits in major crops

Among the 123 reported stress combinations, 58, 41, and 24 were from the abiotic–biotic, abiotic–abiotic, and biotic–

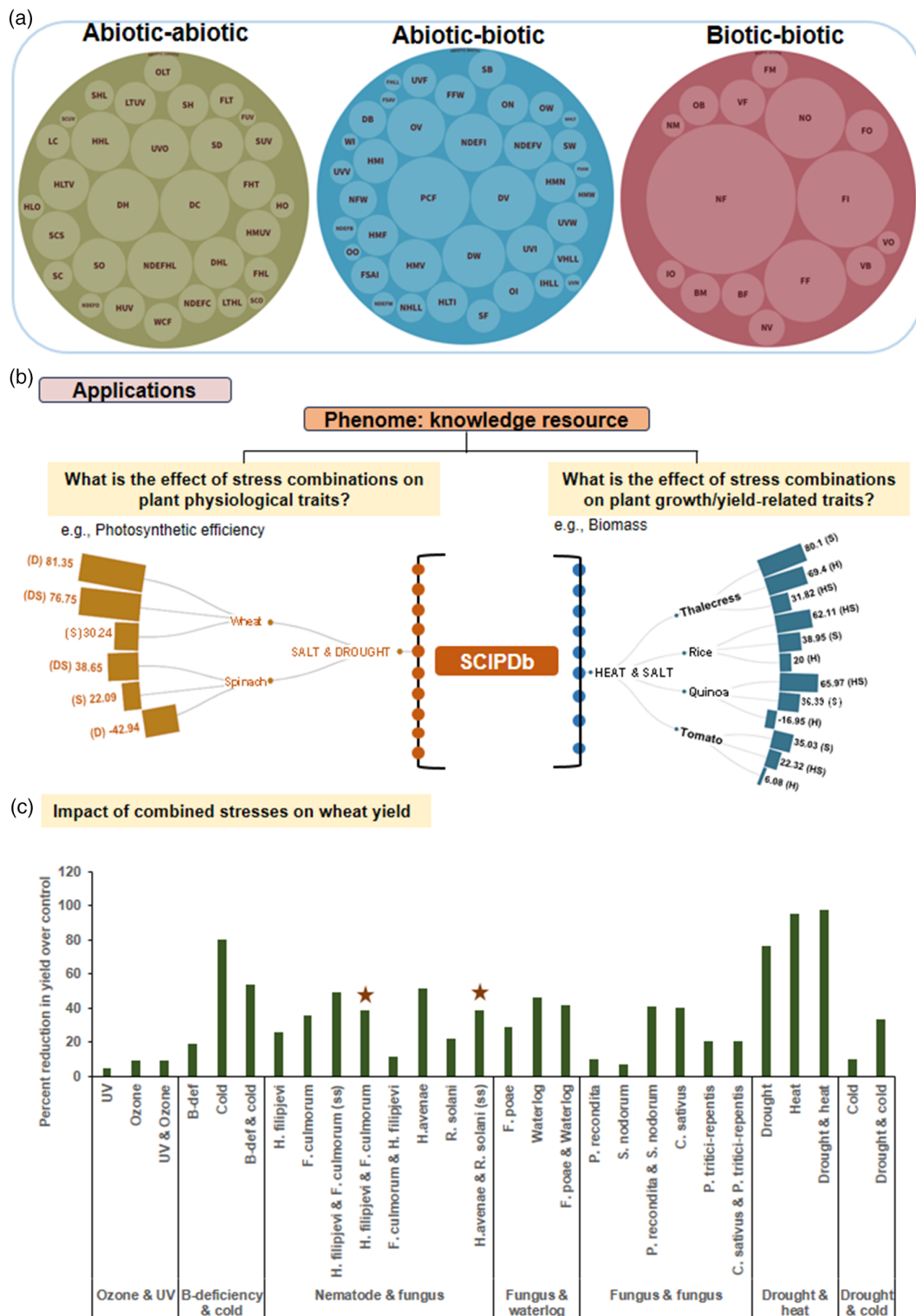
biotic stress categories, respectively (Figure 3a; Figures S8–S10). Of the 58 stress combinations reported in the abiotic–biotic stress category, 87 studies, covering 26 plant species, were of the nematode–fungus stress combination, indicating it as one of the most evident stress combinations (Figure S10, Table S1). Global analysis of yield and yield-attributing traits belonging to plant performance, plant physiological response, and plant pathogenesis response showed greater reductions in yield under the abiotic–abiotic stress category, followed by the biotic–biotic stress category (Figure 3b). Evidently, combined drought and heat stress cause substantial yield losses (Cohen et al., 2021). Several upcoming studies have indicated the role of combined drought, heat, and high light in affecting plant development and metabolism (Zandalinas, Fichman, et al., 2020; Zandalinas, Fritsch, et al., 2020, 2021). Apart from the drought–heat stress combination, fungus–waterlogging and salinity–weeds stress combinations substantially affected the yields of important monocots like wheat and barley, respectively (Figure S11). Wheat yield in particular was greatly affected under drought–heat, drought–cold, boron deficiency–cold, and *Fusarium poae*–waterlogging stress combinations (Figure 3c). However, in the case of nematode–fungus and fungus–fungus stress combinations, the wheat yield response varied with the type of pathogen species involved in the interaction and the order of stress perceived by the plant, as shown in Figure 3c. The database also highlights several other important stress combinations significantly affecting plant yields. For example, in pulses and oilseeds such as peanut, cowpea, soybean, and common bean, yields were more affected under nematode–fungus, ozone–UV, fungus–insects, and drought–weeds stress combinations (Figure S11, Table S1). Among solanaceous crops, fungus (*Verticillium dahlia*) in association with nematodes (*Heterodera rostochiensis*, *Globodera rostochiensis*, and *Pratylenchus neglectus*) showed marked reduction in potato yield (Figure S11). We have also done a comprehensive analysis

Figure 3. Phenome data analysis to assess the effect of stress combinations on agronomic traits.

(a) The bubble diagrams depict the total stress combinations covered in SCIPDb under abiotic–abiotic, abiotic–biotic, and biotic–biotic stress categories. The size of the bubble is directly proportional to the number of studies under the respective stress combinations. For crop-wise stress combinations, bubble diagrams are presented in Supplemental Figures S8–S10.

(b) Schematic representation of the phenome application page and some of the key features offered on various plant traits. Phenome section can be successfully utilized as knowledge resource for data mining and to address pertinent questions in plant science research. We show an example of two questions that can be answered by the database. (i) The database can be used to understand the effect of abiotic–abiotic stress combination on physiological traits, for example, photosynthetic efficiency. The orange dots indicate the ten different stress combinations analyzed in the database. We show an example of heat and salt stress combination, where the bar graphs indicate the percentage reductions in photosynthetic efficiency under the individual and combined stress. (ii) Similarly, SCIPDb can be utilized to assess the impact of stress combinations on plant performance. The blue dots indicate the nine different abiotic–abiotic stress combinations analyzed in the database. We show an example of heat and drought stress combination, where the bar graphs indicate the percentage reductions in biomass under the individual and combined stress. The enlarged versions of radial trees are given in Supplementary Figures S11–S13, and interactive versions are presented in database. Different colored dots are used to signify that the abiotic–abiotic stress combinations analyzed for (i) and (ii) were not same.

(c) As an example for ‘applications’ tab, bar graph shows the impact of various stress combinations on wheat yield. The percentage change in parameter values was computed by comparing the percentage change observed under stress conditions to that of the control. The treatment denoted by a star (*) indicates the application of simultaneous stress, while all other treatments involve sequential stress applications. For abbreviations and pathogen names refer to Dataset S1.



of the effect of different stress combinations on various physiological traits (like photosynthesis, stomatal conductance, photochemical efficiency, Fv/Fm, and chlorophyll content) of various plant species (Figure S12). Further, in

view of understanding the aggravation of plant diseases and emergence of new disease complexes, we found that biotic factors were critical in exacerbating several pathogen infections (Figure S13). In the biotic-biotic stress

category, nematode (*Meloidogyne incognita* and *Heterodera indicus*) and fungus (*Fusarium udum*, *F. oxysporum*, *F. moniliforme*, and *Macrophomina phaseolina*) stress combinations were highly detrimental to maize, pigeon pea, cotton, and chickpea crops, causing higher disease incidence and damage compared to the individual stresses (Figure S13). These results indicate that combined biotic stresses are more detrimental to crops than the stressors individually. In contrast, abiotic stresses have shown a positive effect in terms of reducing pathogen infection and its progression, e.g., ozone–*Phytophthora sojae* in soybean, ozone–*Bean common mosaic virus* in pinto bean, salinity–weeds in sorghum, shade–*Colletotrichum kahawae* in coffee, Mn toxicity–*Uncinula necator* in grapevine, and *Pythium myriotylum*–*R. solani* in peanut showed significant reductions in disease incidence under combined stress (Figure S13). However, recent reviews on combined stress have also indicated that elevated drought, high temperature, and poor nutrient conditions make plants more vulnerable to pest or pathogen infection (Cohen & Leach, 2020; Desaint et al., 2021; Hamann et al., 2020; Savary & Willocquet, 2020).

Global transcriptome changes in combined stressed plants

Transcriptome analysis from 58 combined stress transcriptomes resulted in ~45,169 unique DEGs from 16 plant species ('downloads' section in the database). Functional profiling of significantly enriched DEGs revealed the involvement of genes encoding key proteins like heat-shock proteins (HSPs), Ca²⁺ signal transduction proteins, phytohormone-related genes, defense-related genes, reactive oxygen species (ROS), peroxidases, cell wall-modifying genes, and cytochrome P450 superfamily proteins. Transcription factor (TF) enrichment analysis revealed significant enrichment of dehydration response element-binding protein (DREB), ABA-responsive element-binding protein (ARF), ethylene-responsive element-binding factor (ERF), heat-shock transcription factor (HSF), NAC domain-containing protein, MYB, LOB domain-containing protein, GATA TFs, and WRKY DNA-binding protein families in the DEGs. MYBs and NAC TFs have been reported to regulate pathogen and phytohormone responses like ethylene, jasmonate, and/or salicylate (Bian et al., 2020; Vemanna et al., 2019). MYB TFs have also been reported to regulate the production of secondary metabolites during the induction of stress responses via the phenylpropanoid pathway and cell wall biosynthesis, making them excellent candidates for broad-spectrum stress tolerance improvement in plants (Atkinson et al., 2013; Cao et al., 2020; Rasmussen et al., 2013; Zandalinas, Fritsch, et al., 2020; Zandalinas, Fichman, et al., 2020).

Twenty distinct combined stress transcriptomes were meticulously examined in *Arabidopsis* (Figure 4). This comprehensive analysis unveiled 10 804 DEGs that

exhibited exclusive expression patterns specifically under combined stress conditions. Further stratifying these DEGs into the primary categories of combined stress, followed by an intersection analysis, highlighted 3587, 3182, and 866 DEGs uniquely associated with abiotic-biotic, abiotic-abiotic, and biotic-biotic stress intersections, respectively (Figure 5a).

Conducting pathway enrichment analysis on these DEGs, specific to the major combined stress categories, unveiled a range of pivotal pathway clusters that consistently underwent alteration across the three primary combined stress categories. These encompass pathways integral to amino acid, carbohydrate, energy, carbon, lipid, secondary metabolite, cofactor, and vitamin metabolism (Figure 5b). Furthermore, within the biotic-biotic combined stress categories, pathways related to glycan biosynthesis and key metabolic pathways like glycosphingolipid biosynthesis, glycosaminoglycan degradation, and N-glycan biosynthesis stood out prominently. Unique enrichment was also observed in ethylene and phytochrome signaling pathways, exclusively linked to biotic-biotic combined stress conditions. Intriguingly, under the abiotic-biotic stress category, distinctive interactions were found, including those between sugar and hormone signaling, inositol phosphate metabolism, photosynthesis, and ABC transporter pathways. The abiotic-abiotic stress category exhibited unique pathways such as glycosylphosphatidylinositol (GPI)-anchor biosynthesis, mRNA surveillance pathways, ketone body synthesis and degradation, and glucose sensing and signaling in *Arabidopsis*.

This comprehensive analysis sheds light on the intricate molecular responses orchestrated by *Arabidopsis* under diverse combined stress scenarios, unraveling unique pathway crosstalk and potential key regulators in each stress category.

Deciphering key genes and pathways under combined drought and heat stress by integrative multi-omics

While multi-omics approaches like joint pathway analysis have been limited, they are now being increasingly used in plants (Bjornson et al., 2017; Crandall et al., 2020; Lopez-Hidalgo et al., 2018), with the underlying hypothesis that by combining multi-omics evidence, it will be possible to concretely pinpoint the pathways involved in the underlying biological processes. Carbohydrate metabolism and the expression of related genes have been identified to contribute to the superior heat and drought tolerance of anthers in the rice cultivar N22 compared to the cultivar Moroberekan (Li et al., 2015). The joint pathway analysis approach, integrating changes in gene expression, proteome, and metabolite concentrations in drought and heat combined stress treatments, suggested significant enrichment of four major classes of pathways enriched based on the KEGG BRITE hierarchy. Amino acid metabolism,

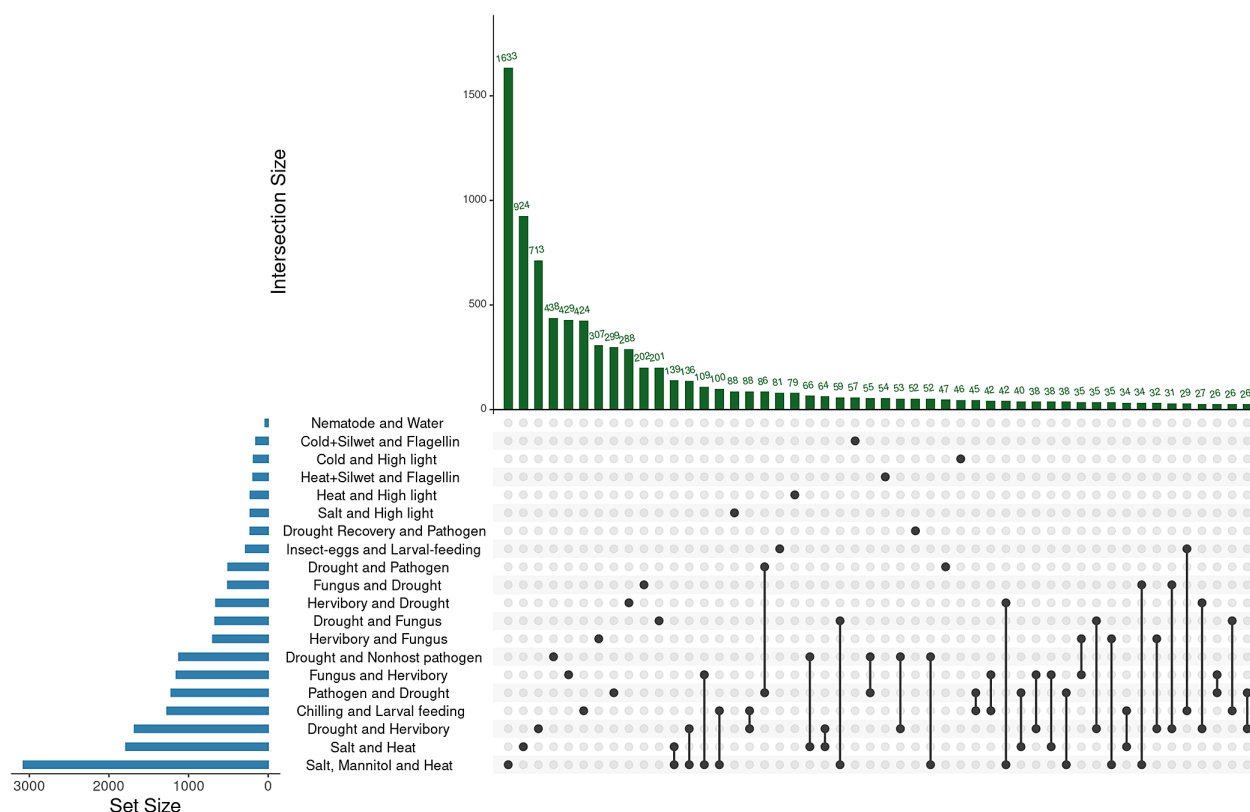


Figure 4. Upset plot depicting intersections of combined stress genes analyzed in multiple combined stress transcriptomes in Arabidopsis. Intersections among combined stress DEGs in Arabidopsis across different stress combinations is presented. The numbers above green bars indicate the number of genes within each intersection. Horizontal blue bars depict set size and set names. Connected dots (black) represent common genes between the transcriptomes, while unconnected dots represent unique genes. Non-overlaps are depicted as gray dots. Data used for the figure is available in Dataset S2.

energy metabolism, carbohydrate metabolism, and signal transduction pathways were supported by all three omics (Figure 6) (Zandalinas et al., 2022). Within the amino acid metabolism class, significantly enriched pathways were related to glutathione metabolism; alanine, aspartate, and glutamate metabolism; glycine, serine, and threonine metabolism; and cysteine and methionine metabolism. Pentose phosphate pathway; glycolysis or gluconeogenesis; and pyruvate, fructose, mannose, ascorbate, aldarate, amino sugar, and nucleotide sugar metabolism pathways were found to be enriched within the energy metabolism class, while carbon fixation in photosynthetic organisms,

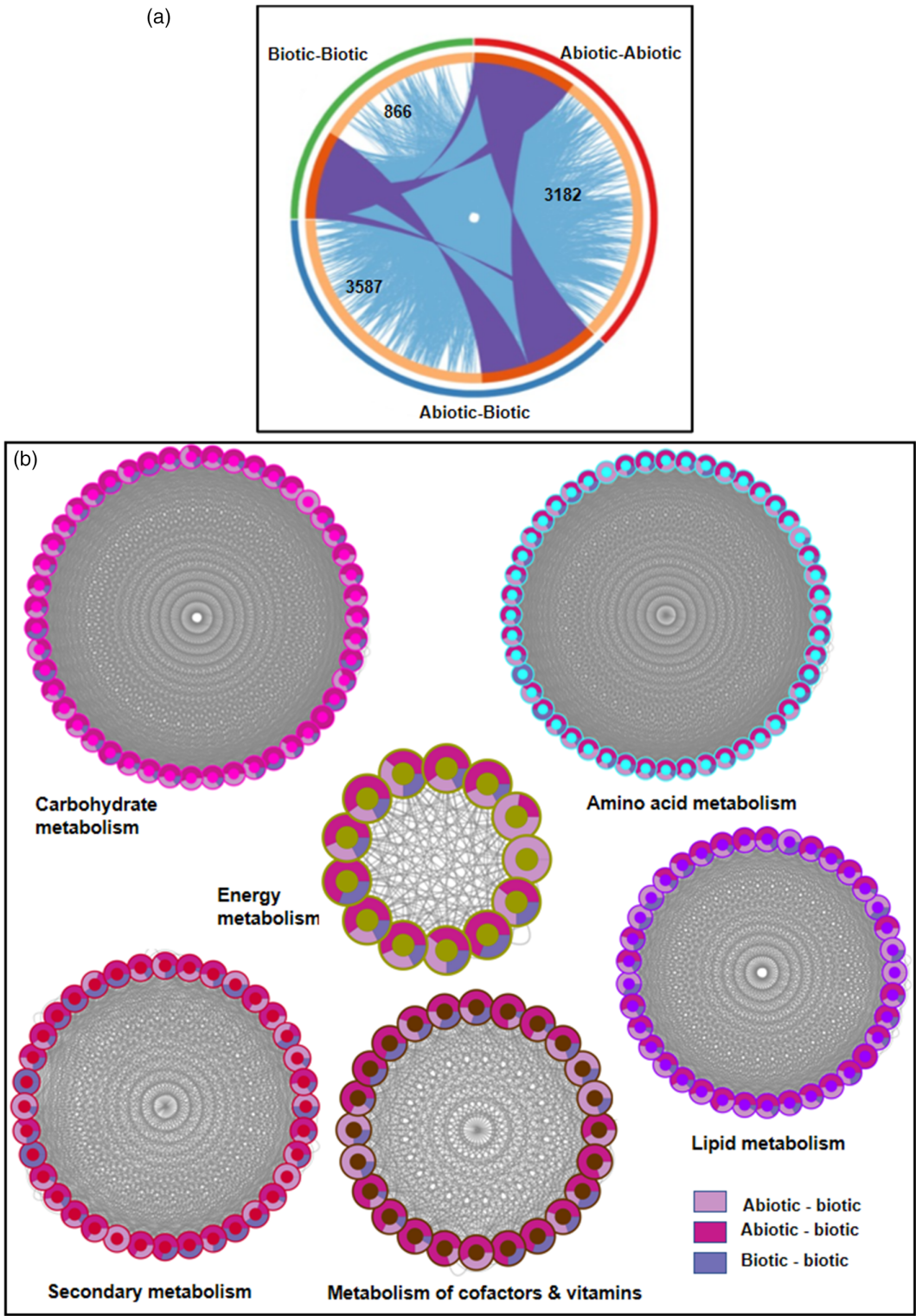
nitrogen metabolism, and sulfur metabolism pathways were enriched under the carbohydrate metabolism class. Among the signal transduction pathways, phosphatidylinositol signaling system pathways mapped to all the three omics data analyzed. Thus, genes commonly associated between these pathway classes may play a significant role in combined stress tolerance in plants (Figure 6).

Applications of SCIPDb and pipeline for novel genes and pathway identification

The applications section of the database provides insight (through multiple case studies) on how to use the diverse

Figure 5. Global pathway and process enrichment analysis of differentially expressed combined stress genes in Arabidopsis.

(a) Circos plot representing the overlap between category-wise DEGs lists in abiotic-abiotic, abiotic-biotic, and biotic-biotic categories. The inner circle represents gene lists, where hits are arranged in the form of an arc. Genes that hit multiple lists are colored in dark orange, and genes unique to a list are shown in light orange. Purple curves link shared genes between the three categories, and blue curves link genes that belong to the same enriched ontology term. (b) Network representation of unique and common pathway clusters among the major combined stress categories. Analysis showed the enrichment of six main pathway clusters, namely, carbohydrate metabolism, amino acid metabolism, lipid metabolism, metabolism of cofactors and vitamins, secondary metabolism, and energy metabolism. Ellipse-shaped nodes depicted as donuts are key pathway clusters (names indicated). The pathway clusters were grouped into broader categories based on KEGG pathway classification (for details on each node, refer to the "Transcriptome – Visualize Transcriptomics data" link in the database). Nodes in the circle represent the genes mapped to those pathways. The color of the nodes indicates the different enriched pathways and their corresponding genes in green. The network is visualized with Cytoscape (v3.8.2) with a "Group by attribute circle" layout. The network of enriched terms is represented as donut charts, where donuts are color-coded based on the identities of gene lists (Dataset S3). The size of a donut is proportional to the total number of hits that fall into that specific term.



datasets hosted in SCIPDb for uncovering the complexities of combined stress in plants. SCIPDb can be used as a knowledge resource to address pertinent unanswered questions

(Harris et al., 2020) in combined stress research, and the information so deciphered can be further used for translational research to discover novel genes and pathways.

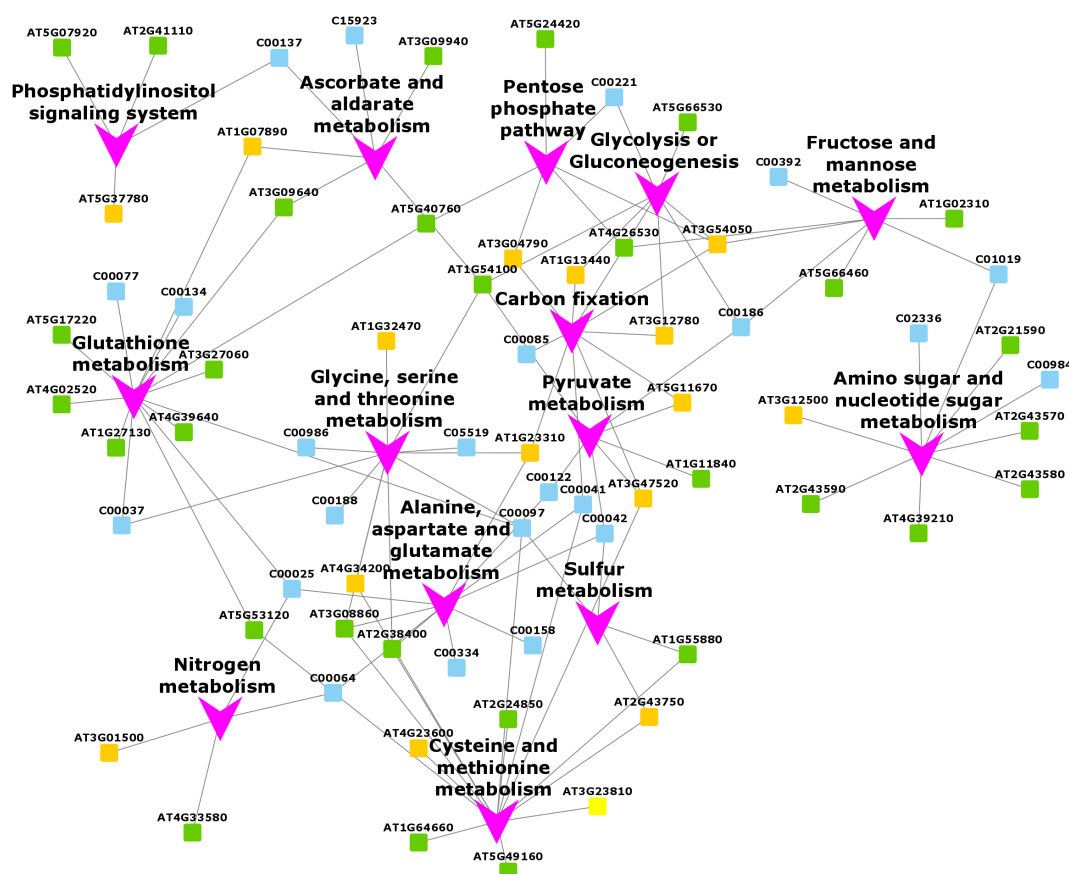


Figure 6. Integrative multi-omics analysis to decipher key omics features and pathways differentially altered during the drought-heat stress combination in Arabidopsis.

The network representation of differentially regulated genes, proteins, and metabolites under the drought and heat stress combination in Arabidopsis was done via joint pathway analysis and visualized in the “y-Files Organic layout” layout in Cytoscape (v3.8.2). The network is presented as nodes indicating various pathways and their associated omics features connected by edges. Edges have been bundled for clarity. The size of a pathway node represents the pathway impact in terms of evidence from omics features, wherein arrow-headed nodes signify pathways having evidence from all the three omics features [transcriptomics (T), proteomics (P), and metabolomics (M)]. Node color corresponds to the class of the pathway or features as mentioned in the node shape and color code box.

SCIPDb as a knowledge resource

Knowledge discovery from databases (KDD) is defined as deciphering valid, novel, potentially useful, and understandable patterns in data. SCIPDb acts as a powerful computational tool for KDD in the area of combined stress research in plants. It has been developed and designed with the aim of extracting new insights from an avalanche of unstructured combined stress data available to date. SCIPDb aims to complement laboratory experimentation by prioritizing candidate genes/pathways, which can significantly accelerate combined stress research (Figures 7 and 8; Table S2).

The extensive datasets presented in the ‘SCIPDb phenotype’ tab help us visualize the impact of different stress combinations on plant growth, yield, and physiology, as well as their interactions with pathogens. Voluminous information has been succinctly depicted as radial trees (Figure 3b; Figures S11–S13). Interestingly, our analysis suggests that the combinatorial infection of nematode and

fungus is one of the most critical stress combinations, apart from the well-known drought–heat and drought–pathogen combinations (Figures S11 and S12). The information compiled in SCIPDb also provides clarity on various inter-microbial interactions affecting plant–pathogen interactions.

From an analysis of 278 research articles, we found that 22 fungi, 13 nematodes, 7 oomycetes, 6 bacteria, and 5 viral pathogens are involved in various biotic–biotic stress combinations. Among them, the most commonly found biotic stressors in disease complexes involve *Meloidogyne incognita*, *Rhizoctonia solani*, and *Fusarium spp.* Similarly, an analysis of 441 research articles on various abiotic–biotic stress combinations revealed drought, salinity, temperature extremes, heavy metals, and low light intensity as the chief abiotic factors affecting plant–pathogen interactions.

Furthermore, SCIPDb hosts an interactive global map illustrating the distribution of various combined stresses

worldwide (Figure 1e). Mapping the geographical distribution of abiotic-biotic combined stresses provides clues as to where environmental conditions exacerbate essential plant diseases. The geographical distribution of

abiotic-abiotic stress combinations, on the other hand, highlights the emerging complex environmental conditions negatively impacting plant performance. Similarly, the global distribution of biotic-biotic stress combinations

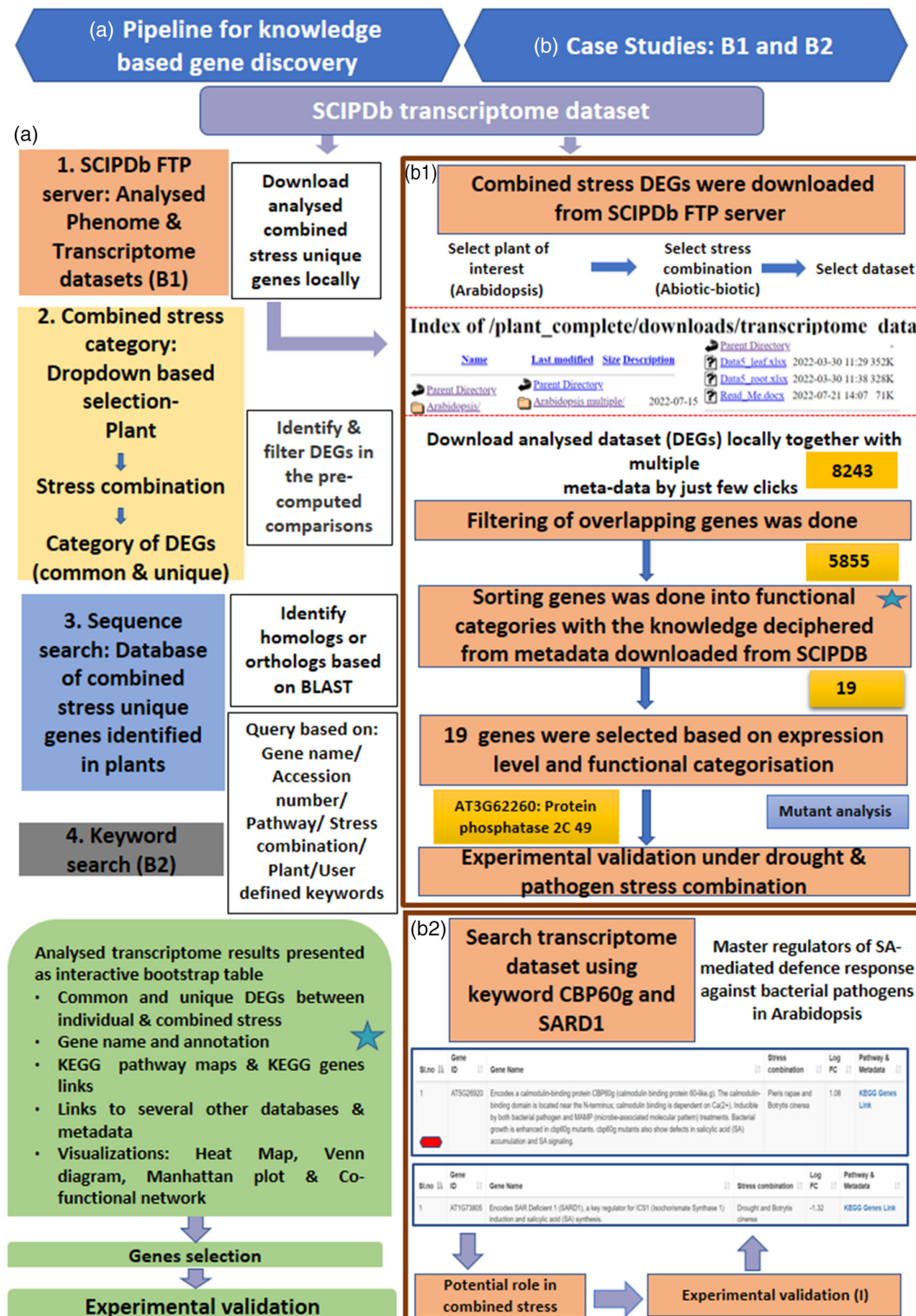


Figure 7. Workflow for identifying key genes responsive to combined abiotic–biotic stress combinations in Arabidopsis using SCIPDb.

(a) Pipeline for knowledge-based gene discovery. The transcriptome datasets hosted in SCIPDb can be mined using four different approaches to filter genes of interest. 1. The SCIPDb FTP server can be used to directly download the analyzed combined stress transcriptomes (unique and shared genes between individual and combined stress treatments). 2. In the combined stress category, it is possible to identify the differentially expressed genes (DEGs) in the pre-computed comparisons based on three-step dropdown-based selection of the plant, followed by the stress combination, and finally, the category of desired genes. It is also possible to visualize the DEGs interactively on KEGG pathways and download several other metadata using the KEGG genes link. 3. SCIPDb hosts a standalone BLAST server and integrated database of unique combined stress genes identified in eight plant species. Users can perform sequence-based searches using the BLAST server in batch mode to identify potential homologs or orthologs. 4. Candidate genes can also be identified using a keyword-based search: The transcriptome search section accepts input in seven categories, namely, gene name, gene ID, pathway, stress combination, plant, and user-defined keywords. The final result is depicted in the interactive bootstrap table, which presents lists of DEGs, gene names and other annotations, expression values, KEGG pathway, and gene information. Multiple interactive visualizations like heatmaps, Venn diagrams, GO enrichment results in the form of Manhattan plots, and co-functional networks, have been provided to aid users in identifying a gene of interest based on functional categorization (marked as a star).

(b) Case studies: Two case studies are illustrated here. b1 depicts a generic pipeline to identify genes of interest in any plant using the SCIPDb FTP server. A five-step workflow is shown starting from downloading combined stress DEGs from the SCIPDb FTP server. This step has been further detailed and demonstrated using screenshots from the SCIPDb. The SCIPDb FTP server was browsed to navigate to the transcriptome datasets section of the database (Path: home/downloads/transcriptome datasets). The server provides a hierarchical organization of analyzed combined stress transcriptomes from multiple plants. The plant of interest (Arabidopsis) and stress combinations were selected (abiotic–biotic). The dataset was selected and downloaded locally. A total of 8243 DEGs from all the abiotic–biotic stress combinations were downloaded, followed by filtering overlapping genes. This step fetched 5855 genes, from which potential genes were selected based on expression values and functional categorization (star indicates use of metadata and visualizations as in A part). b2 demonstrates how the search section of SCIPDb transcriptome was used to identify genes of interest. CBP60g and SARD1 are known regulators of SA-mediated defense responses against bacterial pathogens in Arabidopsis. To decipher their potential role in combined stresses, SCIPDb transcriptome datasets were mined using *CBP60g* and *SARD1*. The results from SCIPDb (depicted in screenshots as red hexagon symbol) showed the potential roles of these genes in combined drought and *Pseudomonas syringae* infection, which was validated by Choudhary and Senthil-Kumar (2022).

provides insight into places where new disease complexes are emerging or being exacerbated by local environmental conditions.

For scenarios where the effect of stressors depended on factors such as stress intensity and order of imposition, interactive scroll bars were employed. These scroll bars allow users to access specific data points by manipulating the scroll bar arrow (Figure S4). A similar presentation of information is also offered in tabular form within the data pages. To synthesize findings from diverse literature and derive overarching conclusions, the study combined data from multiple sources. However, it is important to note that variations in response were expected. This understanding is particularly relevant when interpreting the “Phenomics Data” section, as it may differ from observations made on individual data pages.

SCIPDb as a source for data mining for biological knowledge discovery

Transcriptome datasets hosted in SCIPDb identify large numbers of potential candidate genes that are shared and unique under multiple combined stresses, together with their expression value, expression trend, and several meta-data, including KEGG pathways and gene information. It is often tedious and time consuming for the scientific community to review, curate, and analyze the heterogeneous transcriptome information available from the literature and biological databases in various data formats with divergent quality and coverage. SCIPDb provides ready-to-use, easily interpretable, and downloadable combined stress transcriptomics data for further downstream analysis by end users. A generic pipeline for knowledge-based gene discovery using SCIPDb has been outlined in Figure 7a and is also hosted in the “Application” tab of the database. The

figure outlines four different ways to mine analyzed transcriptome datasets hosted in SCIPDb to filter genes of interest.

SCIPDb for biological knowledge discovery for gene prioritization

Multiple visualizations and computational tools hosted in SCIPDb can aid in the filtering of genes based on a user’s prior knowledge and interest. This can help in prioritizing candidate genes and pathways that when perturbed through potential experimental interventions, can have a desired biological outcome in plants. Interactive visualizations such as heatmaps, Venn diagrams, and GO enrichment results in the form of Manhattan plots and co-functional networks have been provided to aid users to further filter and identify genes of interest based on functional categorization (Figure 7a). Further SCIPDb transcriptome dataset and its associated data and visualizations can be used to mine, filter and identify candidate genes involved in defense against stress combinations using a generic pipeline described in Figure 7b. This figure demonstrates how CBP60g and SARD1, which are known regulators of SA-mediated defense responses against bacterial pathogens in Arabidopsis, were identified (Choudhary & Senthil-Kumar, 2022). Although transcriptomic analysis of various combined stressed plants has been done (Cohen et al., 2021; Prasad & Sonnewald, 2013; Zandalinas, Fichman, et al., 2020; Zandalinas, Fritsch, et al., 2020), SCIPDb presents a comparison of transcriptional changes occurring due to different stress combinations. Using the pipeline, we identified putative candidate genes involved in defense against drought and pathogen stress in Arabidopsis and chickpea (Table S2). The pipeline can also be applied to identify combined stress-related genes in other plant

species as per the user's interest. From our gene list consisting of seven genes, we found *AtPP2C49* in Arabidopsis as an interesting novel candidate gene under drought and pathogen stress (Figure 8). Protein phosphatase 2C (PP2Cs) are class protein known to regulate ABA signaling pathway through SnRK2 function (Miyazono et al., 2009). In total 19 PP2Cs have been identified in Arabidopsis and some are involved in plant immunity against different pathogens (Bhaskara et al., 2019; DeFalco et al., 2022; Mine et al., 2017).

Omics feature and novel pathways identification

SCIPDb has been used to identify novel combined stress-responsive pathways and omics features for abiotic-biotic stress combinations in Arabidopsis (Figure 9). We integrated the transcriptome datasets downloaded from SCIPDb with the metabolome datasets using the integrative joint pathway analysis approach. The major pathways identified were categorized into amino acid metabolism, carbohydrate metabolism, energy metabolism, lipid metabolism, and signal transduction based on KEGG BRITE hierarchy (Dataset S4). Finally, alanine, aspartate, and glutamate metabolism were chosen from the amino acid metabolism category (one of the top pathways enriched), and omics features (transcripts and metabolites) mapped onto it were back searched using the search button of SCIPDb to decipher their potential role under combined stress. Based on the result obtained from SCIPDb, we propose alanine, aspartate and glutamate metabolism as a novel pathway [through asparagine synthetase 1 (*AtASN1*), AT5G65010 in Arabidopsis] with a potential role in defense against drought and pathogen stress combination.

FUTURE PERSPECTIVES

SCIPDb is a comprehensive database amenable to data mining and data-driven research on combined stresses in plants. Global phenome data analysis shows that abiotic-biotic stress combinations are major threats to crop productivity. In the face of global climate change, the occurrence of these stress combinations is projected to increase in coming years. Therefore, dedicated studies on this aspect are essential to sustain crop yields in the future. Key takeaways from yield analyses are that monocots will be more affected under the abiotic-biotic and abiotic-biotic stress category, whereas pulses, oilseeds, and vegetable crops will be more affected under the biotic-biotic stress category.

Our unique combined stress integretome developed using multi-omics data integration highlights sugar metabolism, energy metabolism, and amino acid metabolism as the key pathways operating under combined stress conditions. The addition of proteomics and metabolomics data to the database with multi-omics analysis will further

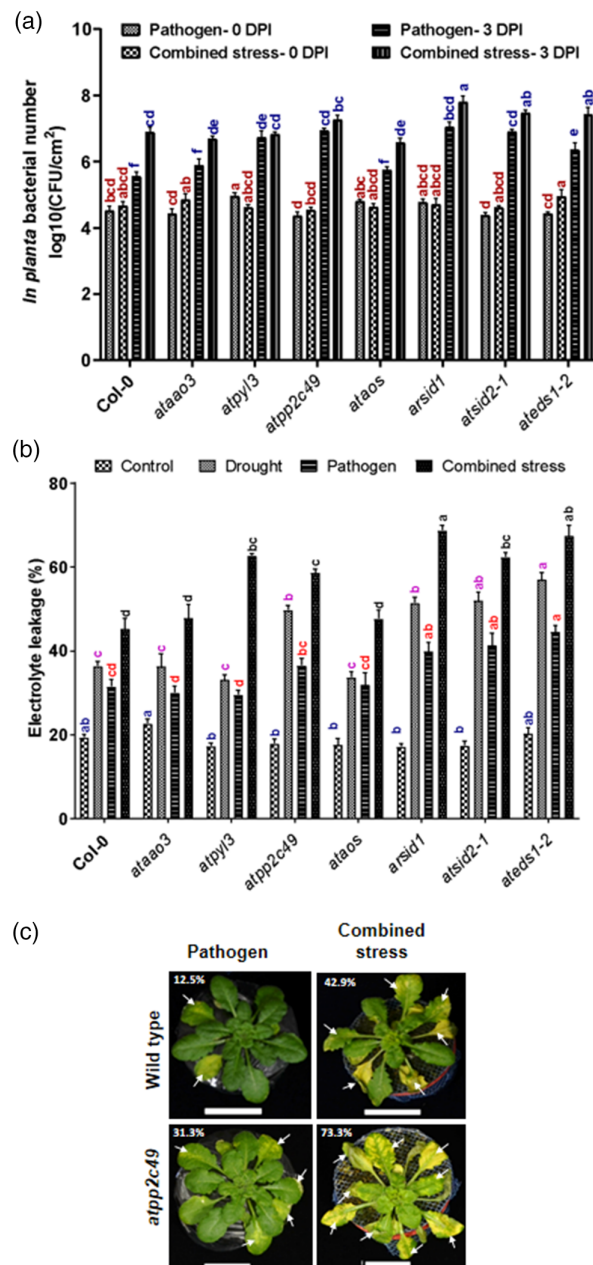


Figure 8. SCIPDb as a tool for identifying candidate genes involved in combined abiotic-biotic stress resistance: Evidence from drought and pathogen studies in Arabidopsis.

Out of several genes shortlisted from the SCIPDb transcriptome (as described in Figure 7) seven were validated for their role in defense against combined drought and *Pseudomonas syringae* pv. *tomato* DC3000 (Pto DC3000) infection in *Arabidopsis thaliana*. Graph showing (a) *in planta* bacterial number and (b) electrolyte leakage in mutants under combined drought and Pto DC3000 infection. Bars represent data collected from at least six biological replicates. Error bars designate the standard error of mean, and different letter indicates the significant difference at $P < 0.05$ (two-way-ANOVA followed by Duncan's multiple range test). The experiment was repeated thrice and found similar trend.

(c) Plant phenotype showing the extent of disease incidence in wild-type and *atpp2c49* mutants under pathogen-only and combined stress treatments. Numbers inside the pictures indicate the percent disease incidence under respective treatments.

demystify combined stress responses of plants. Unraveling the mechanisms by which the molecular signatures associated with these pathways impact plant responses to combined stress can open new vistas for developing resilient crop varieties with better adaption to changing climate and global warming.

The pipelines presented in the database can be useful to identify novel genes and pathways from other crop and model plants for various stress combinations. Hence, this database can be used to advance fundamental knowledge of plant stress tolerance mechanisms. Further, we have shown case studies in this manuscript for substantiating the same along with the potential for applying the information to improve plant performance under stress.

With the continual accumulation of available data in the field of combined stress, we will update the database annually by incorporating newer studies. There have been recent advancements on the varying vulnerability of different growth stages and plant tissues to combined stresses. Considering this, based on data availability we will be segregating the transcriptomic datasets based on growth stages and plant tissues. We intend to add other omics datasets related to combined stress research in the future version of SCIPDb, together with novel features like prediction modeling based on machine learning and meteorological data integration with geographical distribution information. Overall, SCIPDb is an informative and valuable resource for combined stress research in plants.

MATERIALS AND METHODS

Bio curation and integration of phenome datasets into SCIPDb

Literature mining

To retrieve all available articles and to have greater than 90% literature coverage, several search engines were queried using suitable and carefully designed keywords (including several variants). Bibliography from each article was also searched to achieve better coverage. Table S3 details the search engines used for the phenome literature mining. A detailed pipeline is presented under the SCIPDb methodology tab (http://223.31.159.3/plant_complete/methlogy_orangesunset.php).

Sorting of articles

After retrieving all the articles related to particular stress combination, they were sorted as 'main research articles (Science Citation Index articles)' and 'ancillary articles'. This was done based on the type of articles (e.g., research, review, reports, etc.) and type of data (e.g., morpho-physiological and molecular data) present in the articles. Main research articles with only morpho-physiological data were considered for data extraction, whereas reports, thesis, book chapters, abstracts, reviews, gene overexpression, and gene-silencing studies were listed under ancillary articles and not considered for data extraction. All these types of articles were integrated into the database under the 'phenomics' tab. Plant competition, in this database, implies intra-specific

competition and constituted studies related to the effect of plant density involved with other stresses on plants. Cases of interspecific competitions, in most of the cases, have not been discussed under this category. The studies involving plant growth promoting bacteria and fungi were excluded as both of them independently do not act as pathogen to plants. In most cases tree species were not included, however, in some places where the economically important tree species have been considered. Studies wherein the experiments involved young saplings were included. The studies with gene overexpression or silencing were not considered for the data extraction however they were included as ancillary articles under the particular stress combination. Articles with chemical treatment like fungicide, herbicide, and fertilizer amendments were also excluded since these are anthropogenic stresses.

Listing out parameters and their classification

From the main research articles, parameters studied in each article were listed out and classified into type A, B, and C parameters based on their significance in reflecting the net impact of the stress. **Type A** includes growth (plant height, biomass, leaf area, leaf number, root length, shoot weight, root weight, etc.) and yield (seed weight, seed number, test weight, etc.), attributing parameters that directly reflect the impact of stress, while **Type B** includes physiological (photosynthesis, stomatal conductance, transpiration, chlorophyll content, etc.) and pathogenesis (disease index, pathogen load, disease score, etc.) related parameters which indirectly reflect the impact of stress. Finally, biochemical parameters such as proline content, MDA content, nutrient composition, ROS content, etc., which also explains the impact of stress but to a lesser extent compared to the other two classes of parameters were categorized as **Type C**. For the complete list of parameters, refer "Traits" Tab in the SCIPDb (http://223.31.159.3/plant_complete/traits.php).

Data extraction and depiction

Once parameters were listed out from each article, data values were extracted into the Excel file. The values from the table were directly copied into the Excel sheet, whereas values from graphs were extracted using the 'GetData Graph Digitizer' (<http://getdata-graph-digitizer.com/>) tool for better accuracy. To make the heterogeneous data uniform, normalization was done using the formula mentioned below.

$$\text{Change over control (\%)} = \frac{(\text{Control} - \text{stress}) * 100}{\text{Control}}$$

Few parameters, such as electrolyte leakage, pathogenesis-related parameters, etc., were not subjected to calculation. Using both the calculated and un-calculated values, a table was prepared, reflecting the net impact of stress and interaction between the two stresses at the plant interface. This table was used for preparing the 'data page' file for each study on a specific plant and was finally represented in the database in tabular form. For easy understanding, percent change values were shown along with arrows in red and green color. A red-colored downward arrow indicates that the parameter is affected under stress; the higher the positive value greater the damage to the parameter under stress. Green-colored upward arrow indicated parameters were not affected under stress conditions as compared to control.

Data analysis and interpretation

Data presented in tabular form was analyzed by comparing the individual and combined stress values of each parameter. When the percent change values was greater in combined stress

compared to both the individual stresses, the outcome of combined stress was depicted as 'negative'. If percent change values were less in combined stress compared to both the individual stress, then the outcome was depicted as 'positive'. Together with the tabular part, a brief inference was also documented for each article. Each of the data pages has a brief introduction of the stress combination, with information about the number of studies available in different crops for that particular stress combination. Mapping of important traits hosted in SCIPDb was done using the Plant Trait Ontology (<http://www.ontobee.org/ontology/TO>) and Planteome (<https://planteome.org/>) databases wherever possible.

The impact of stress combinations was evaluated considering the developmental stage of the plants, the timing of stress application, and the order in which stressors were introduced. Detailed data pages were constructed to present information for each study conducted. An illustrative example can be found on the data pages where the top right corner showcases the factors under consideration. Each study was treated independently, and analyses were conducted on an individual basis, avoiding data aggregation. Acknowledging potential variability in stress response across different genotypes, distinct links were provided on the data pages to download genotype-specific Excel sheets containing comprehensive data.

The methodology employed for this study included a comprehensive consideration of influencing factors, individualized study analyses, inclusion of genotype-specific responses, interactive data exploration tools, and recognition of potential variations in the aggregated data analysis.

Phenomics datasets integration into the SCIPDb

The frontend user interface was implemented using HTML5, CSS and PHP (version: 7.0.12). The dataset was finally integrated and depicted as an HTML page presented to the users based on the three-level-dropdown selection designed in JavaScript, specific to each plant species. The back-end schema was designed using MySQL, an open-source relational database management system, and stored in MySQL tables (Version: 5.7.17). To provide an interactive interface and enhanced user experience, Bootstrap 4 and jQuery were used. For data page format, please visit the 'phenomics' section of database (http://223.31.159.3/plant_complete/data.php).

Visualization of phenomics dataset

We employed Tableau Public Desktop (Version 2020.4, available at <https://public.tableau.com/en-us/s/>) to visualize intricate phenomics data with high dimensions. This visualization took the

form of an interactive Treemap/stress matrix. The stress matrix was constructed based on the frequency of studies demonstrating positive or negative outcomes within various stress combinations. In this context, a positive interaction denoted that the combined stress resulted in less harm to the plants compared to individual stresses. Conversely, a negative interaction indicated instances where the plants experienced greater damage under combined stress conditions. To illustrate, if a particular stress combination was the subject of 12 research articles and the majority of these articles reported a favorable impact, the combination was categorized as "positive." Equally distributed instances of positive and negative outcomes led to a classification of "positive/negative."

For the development of radial trees, we utilized Flourish Studio (accessible at <https://app.flourish.studio/>). Additionally, we harnessed various interactive visualization tools offered by Flourish Studio, including Chord diagrams and Sankey Diagrams, to effectively represent the analyzed information. Furthermore, an interactive geographical map was generated using Google My Maps, which can be accessed at <https://www.google.com/maps/about/mymaps/4.2>.

Bio curation and integration of transcriptome datasets into SCIPDb

Data mining

The relevant transcriptome datasets for combined stress in plants were compiled and curated using two major public databanks for microarray data, including Gene Expression Omnibus (GEO) (<https://www.ncbi.nlm.nih.gov/geo/>) and Array Express (<http://www.ebi.ac.uk/arrayexpress/>). The NCBI GEO and ArrayExpress functional genomics repository were queried using keywords, "combined stress" AND "Plants" [organism]. For the compilation of RNA-seq transcriptomics data NCBI, Sequence Read Archive (SRA) (<https://www.ncbi.nlm.nih.gov/sra>) database was used.

Data curation

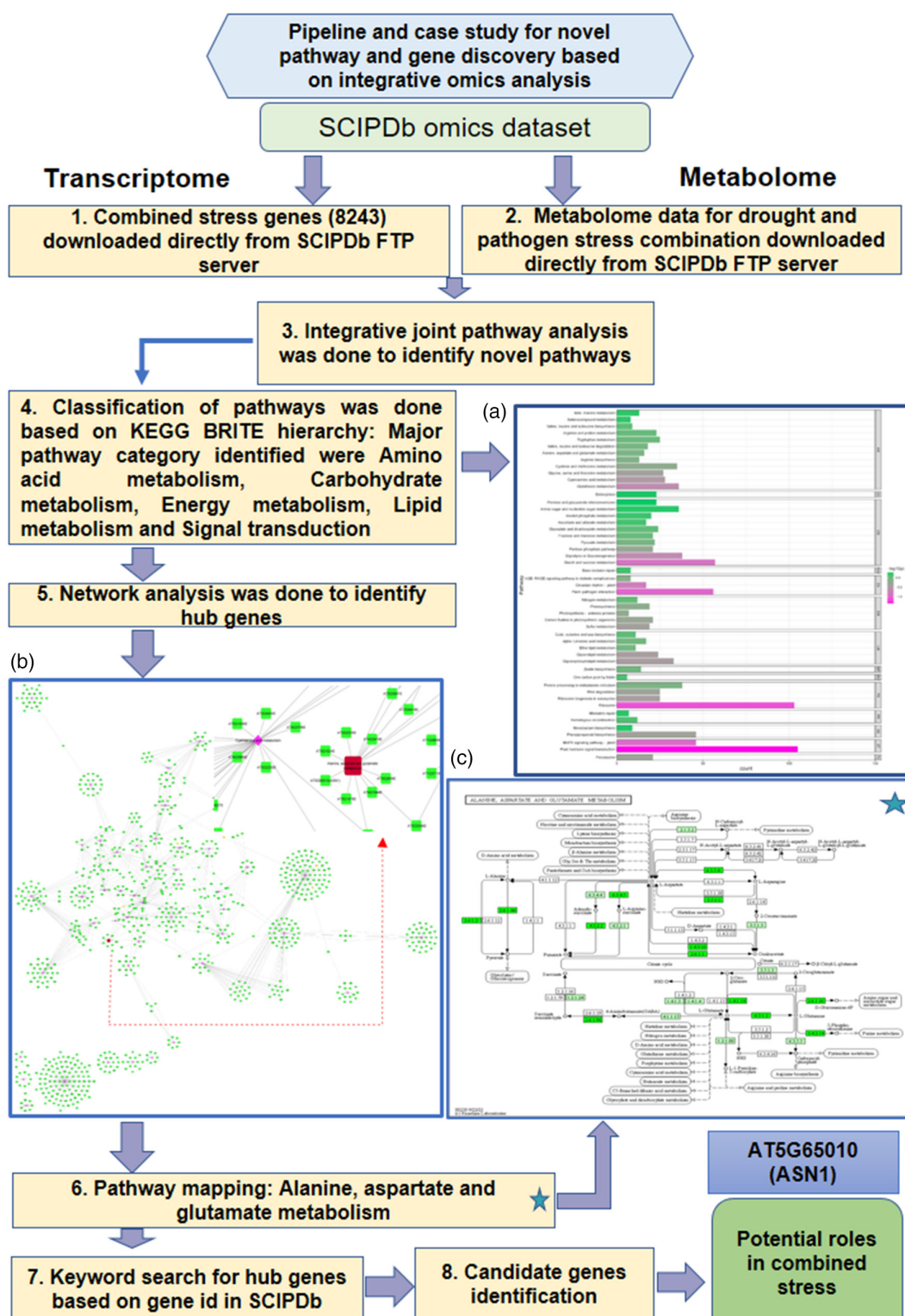
The transcriptomic datasets mined from public databases were manually curated, to determine whether they were from actual combined stress studies. Also, duplicate studies were filtered out. For the final analysis, studies that showed availability of completely raw and processed datasets and also at least 4 samples (2 combined stress treatments and 2 respective controls, comparable case and control samples) were included.

Analysis of transcriptomics dataset

Microarray data mined from relevant studies were analyzed in the R environment using the *GEOquery* (<http://www.bioconductor.org/>)

Figure 9. Workflow for unveiling novel pathways and omics features in response to abiotic and biotic stress combinations in Arabidopsis.

1. The transcriptome datasets hosted in SCIPDb were mined using the in-house FTP server to download and filtered all the DEGs from abiotic-biotic stress combinations. The four-step workflow (Figure 7) was used to download 8243 genes. 2. Similarly, the analyzed metabolome dataset was downloaded for the drought and pathogen stress combination. 3. Integrative joint pathway analysis was done using MetaboAnalystR to identify novel pathways and omics features (transcripts and metabolites) differentially altered under drought and pathogen stress combinations. 4. The pathways identified were further classified using KEGG BRITE hierarchy into major pathway categories. The top 50 pathways (sorted based on Q value) plotted in the form of a bar graph are shown in (a). The Y-axis depicts the pathway names, while the X-axis shows the gene count (number of genes or metabolites mapped onto these pathways). Bar color is based on the $-\log_{10}(P)$ value. The visualization has been faceted based on major KEGG pathway categories, namely, AM: amino acid metabolism, CM: carbohydrate metabolism, LM: lipid metabolism, EM: energy metabolism, ST: signal transduction, PS: protein synthesis, EA: environmental adaptation, TC: transport and catabolism, SS: secondary metabolite synthesis, MP: metabolism of terpenoids and polyketides, RR: replication and repair, MV: metabolism of cofactors and vitamins, DNAR: DNA replication and repair, and CC: cellular transport and catabolism. AM, CM, EM, LM, and ST were the main pathway categories identified. 5. Network analysis was done to identify major gene clusters or hub genes. The top 50 pathways (magenta diamond-shaped nodes) and their associated omics features (transcript: green square-shaped nodes, metabolites: yellow octagon-shaped nodes) are shown in (b). 6. Alanine, aspartate, and glutamate metabolism were chosen from the amino acid metabolism category [highlighted in maroon in (b)] and omics features were mapped onto it (shown in magnified view). The mapped omics features are represented in (c) in green. 7. To further gain insights into the functional role of all the mapped omics features, they were searched using the keywords (gene names) in SCIPDb. 8. Asparagine synthetase 1 (ASN1, AT5G65010) showed a potential role under combined stress.



packages/2.8/bioc/html/GEOquery.html) and limma (Linear Models for Microarray Analysis) (<http://www.bioconductor.org/packages/release/bioc/html/limma.html>) R packages from the Bioconductor project. The GEOquery R package performs the initial parsing of

the GEO data into R data structures. This data is used by the limma R package for identifying differentially expressed genes (DEGs) in the input dataset. Following this probe ID conversion was done for DEGs obtained from individual microarray datasets.

Based on the platform used in the individual studies probe ID match was done to convert the DEGs into a common ID format. Affymetrix (https://www.affymetrix.com/analysis/netaffx/xmlquery_ex.affx?netaffx=wtgene_transcript) platform was used to do the probe ID conversion to fetch their corresponding transcript IDs. For Agilent and other platforms, the bioDBnet (biological DataBase network) platform was used (<https://biodbnet-abcc.ncifcrf.gov/db/db2db.php>). In cases where multiple probes matched the same locus, the probe ID with the highest fold change was considered. For RNA-seq transcriptome datasets analysis, ENA (European Nucleotide Archive) (<https://www.ebi.ac.uk/ena/browser/home>) and SRA (Sequence Read Archive) toolkit (Version 2.10.8) were used to download and split the SRA raw read data from NCBI (National Center for Biotechnology Information) into FASTQ files (<https://www.ncbi.nlm.nih.gov/>). Raw sequence reads in the format FASTQ files, thus downloaded were subjected to quality check by FastQC (<https://www.bioinformatics.babraham.ac.uk/projects/fastqc/>) followed by removal of adapter sequences and subsequent trimming of the reads using Trimmomatic (<http://www.usadellab.org/cms/?page=trimmomatic>) or Cutadapt (<https://cutadapt.readthedocs.io/en/stable/>) based on the FASTQC report. This was followed by mapping the reads to the reference genome using HISAT2 (Hierarchical Indexing for Spliced Alignment of Transcripts 2, version 2.0.1) (<http://www.ccb.jhu.edu/software/hisat/index.shtml>) which is a fast and sensitive spliced alignment program for mapping RNA-seq reads. SAM (Sequence Alignment/Map format) files to BAM (Binary Alignment/Map format) files conversion was done using SAMtools (<http://samtools.sourceforge.net>, version 0.1.19). BAM files were then further sorted. Subsequent to this counting of reads mapped to individual genes or transcripts was done using bedtools (<https://bedtools.readthedocs.io/en/latest/content/installation.html>). In the final step differentially expressed genes between different combined stress and individual stress treatments were identified using DESeq2 in R (<https://bioconductor.org/packages/release/bioc/html/DESeq2.html>). For plants like *Solanum dulcamara* where the reference genome wasn't available de novo RNA Seq analysis was performed. After quality check and subsequent trimming of reads de novo assembly was generated using Trinity (<https://github.com/trinityrnaseq/trinityrnaseq.github.io/blob/master/deprecated/index.asciidoc>) followed by abundance calculation using RSEM (<https://github.com/deweylab/RSEM>) or Salmon (<https://salmon.readthedocs.io/en/latest/salmon.html>). Differential expression analysis was done using Trinity-Edge R (<https://bioconductor.org/packages/release/bioc/html/edgeR.html>). Assembled reads were annotated using the NR/NT NCBI blast database (<https://blast.ncbi.nlm.nih.gov/Blast.cgi>). For transcriptome studies where analyzed data was already provided by the authors, DEGs were directly downloaded and used for further analysis pipelines.

Biological interpretation of transcriptome data

The DEGs identified under each stress combination were associated with several meta-data, like, gene names and mapped to 'KEGG (Kyoto Encyclopedia of Genes and Genomes) Pathway'. In order to aid users to navigate through several other databases and cross-references outside KEGG and fetch a deluge of meta-data corresponding to a particular entry, KEGG genes links were provided in cases where pathway information was not available. KEGG API (Application Programming Interface) (<https://www.kegg.jp/kegg/rest/keggapi.html>) was used to fetch both of these links for respective DEGs. In order to further gain insights into the biological functions of DEGs identified by transcriptome analysis, functional and pathway enrichment analysis was performed using the gProfileR package (<https://cran.r-project.org/>).

Benjamini–Hochberg adjustment for multiple hypothesis testing was used to correct the *P*-values. The over-represented GO (Gene Ontology) and KO (KEGG Ontology) terms measured by the adjusted *P*-values depicted gene function and biological pathway associations.

Data integration into the SCIPDb

The frontend user interface was implemented using HTML5, CSS, JQuery and PHP (version: 7.0.12). HTML pages were designed to accept queries based on a three-level dropdown-based selection using JavaScript, specific to each plant species. The back-end schema was designed using MySQL, an open-source relational database management system, and data was stored in mysql tables (version: 5.7.17). To provide an interactive interface and enhanced user experience, we used Bootstrap 4, to present final results. For further details please visit the 'transcriptomics' section of database (http://223.31.159.3/plant_complete/transcriptome.php).

Visualizations

To visualize the high-dimensional transcriptomics data, several in-house scripts were generated using shell scripts and R. The heat map was made interactive using the "Heatmaply" (<https://github.com/talgali/heatmaply>) package from R. "VennDiagram" (<https://cran.r-project.org/web/packages/VennDiagram/>) package was used and scripts were customized to generate color-coded Venn diagrams representing specific categories. Gene ontology enrichment analysis and interactive visualization of the results in the form of Manhattan plots were done using "gprofilerR" package (<https://cran.r-project.org/web/packages/gProfileR/index.html>). Correlates data for the top twenty genes in each category have been collected from ATTED-II (<http://atted.jp/>) which is a plant co-expression database and for Arabidopsis from AraNetv2 (<https://www.inetbio.org/aranet/>). Functional annotation and pathway mapping for the correlates was done using KEGG and TAIR. Final visualization of the network together with the other meta-data was done using Cytoscape (<https://cytoscape.org/>), an open-source software platform, widely used for visualizing complex networks. Transcription factor enrichment analysis was done using EatupTF tool (<http://chromatindynamics.snu.ac.kr:8080/EatupTF>).

In the process of creating co-functional gene networks for *A. thaliana* and other plants, we utilized AraNet v2, accessible at (<http://www.inetbio.org/aranet>). Correlated genes associated with the top 20 uniquely differentially expressed genes under combined stress were retrieved from the AraNet server, each interaction being assigned a log-likelihood score (LLS) reflecting the probability of genuine functional linkage between genes. For visualization purposes, the downloaded data underwent conversion into a sif format (Simple Interaction Format) using the Cytoscape.js library. The resultant visualization focused on the top 200 correlated genes, while the full correlation dataset remains available for reference. A Node Attributes (.noa) file was compiled by merging various datasets, offering users comprehensive information accessible through interactions with individual nodes. This includes details such as Gene Ontology Biological Process (GO-BP), Gene Ontology Molecular Function (GO-MF), Gene Ontology Cell Compartmentalization (GO-CC), log-likelihood scores (LLS), rank among the top 200 correlated genes, evidence codes, paralog information, linked queries, candidate genes, and gene symbols. Moreover, edge information, attained by clicking on network edges, provides insights into connected nodes. The visualization process was empowered by the Cytoscape.js library, an open-source JavaScript-based graph theory and network library, allowing interactive graph manipulation and analysis. A range of

interactive features, such as pinch-to-zoom, box selection, and panning, were supported, along with customizable styling, layouts, and background options. Ultimately, this approach enriches the exploration of DEGs by unveiling their functional relationships amid distinct stress conditions.

Arabidopsis combined stress transcriptome

Upset plot was generated using UpSetR package, while circos plot was generated using Metascape, a gene annotation & analysis resource. (<https://metascape.org/gp/index.html#/main/step1>). Pathway enrichment analysis was done using major pathway databases like KEGG (<https://www.kegg.jp/kegg/rest/keggapi.html>), Aracyc (<https://plantcyt.org/typeofpublication/aracyc>) and Wiki-pathways (<https://www.wikipathways.org/index.php/WikiPathways>). Final visualization and network analysis were done using Cytoscape, v3.8.2 (<https://cytoscape.org/>).

Integretome analysis

MetaboAnalystR package was used to perform joint pathway analysis of transcriptome, proteome, and metabolome profiles. For enrichment analysis (ORA) hypergeometric analysis was used, while for topology measure, degree centrality was used. Combining *P*-values at pathway level was used for integration of three omics datasets. Final visualization and network analysis was done using Cytoscape, v3.8.2 (<https://cytoscape.org/>). A detailed list of enriched pathways is given in dataset S3.

FUNDING INFORMATION

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AUTHOR CONTRIBUTIONS

MS-K conceived the idea, designed the database, outlined the manuscript, and provided all resources. Pi.P. developed the webtool, performed data integration and data visualization. MP and Pr.P. contributed to the phenomics part. Pi.P, MP, and Pr.P. performed overall data analysis. AS contributed to the data collection for the phenomics part of the manuscript. Pi.P. and VSB contributed to the data collection and analysis part of transcriptomics. MS-K, Pi.P, MP, and Pr.P. drafted the manuscript. MS-K edited and finalized the manuscript and SCIP database. All authors agreed to the submitted version of the manuscript.

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CONFLICT OF INTEREST

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The curated phenomics, genotypes, transcriptomics, and references are hosted on a local FTP server and can be downloaded by the users using the link http://223.31.159.3/plant_complete/downloads.php.

SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article.

Dataset S1. Raw data for phenome data analysis to assess the effect of stress combinations on agronomic traits.

Dataset S2. DEGs under various stress combinations in *Arabidopsis thaliana* used to generate Upset Plot

Dataset S3. GO and Pathway enrichment analysis of DEGs under various stress combination in *Arabidopsis*

Dataset S4. Integrative multi-omics analysis to decipher pathways jointly supported by all omics features (transcriptome, proteome and metabolome) during drought and heat stress combination in *Arabidopsis*.

Figure S1. Illustration indicating the various effects of abiotic stresses on phytopathogens and the plant defense

Figure S2. Content and construction of SCIPDb.

Figure S3. Combined stress transcriptome analyzed and integrated into SCIPDb.

Figure S4. A typical data page entry for phenome in SCIPDb.

Figure S5. Stress matrix representing the impact of stress combinations curated and integrated in SCIPDb.

Figure S6. Literature analysis of combined stress articles published from 1950 to 2021.

Figure S7. A typical data page entry for transcriptome in SCIPDb and its associated visualizations.

Figure S8. Literature analysis of combined stress articles published from 1950 to 2021 under the abiotic–abiotic stress category.

Figure S9. Literature analysis of combined stress articles published from 1950 to 2021 under the abiotic–biotic stress category.

Figure S10. Literature analysis of combined stress articles published from 1950 to 2021 under the biotic–biotic stress category.

Figure S11. Literature analysis of combined stress growth and yield data for various plant species.

Figure S12. Literature analysis of combined stress physiological data for various plant species.

Figure S13. Literature analysis of disease incidence data under combined stress for various plant species.

Table S1. List of references of important stress combinations discussed in results section of manuscript.

Table S2. List of putative candidate genes identified from SCIPDb with potential role under combined drought and pathogen stress.

Table S3. List of search engines used for literature mining for Phenomics dataset.

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